

Network Security

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ES ZG513/ SS ZG513, Network Security
Lecture No VII.

Public Key Cryptography

- Each user has public and private keys
- Public key used for encryption
- Private key used for decryption
- Solves key distribution problem

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Reference Model

- Key Generation: produces public and private keys
- Encryption: uses public key
- Decryption: uses private key
- Security relies on computational hardness

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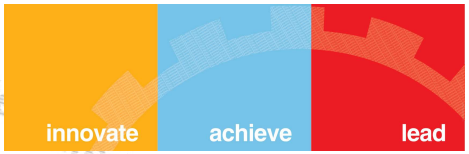
Mathematical Foundations

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- Modular arithmetic
- Prime numbers and Euler totient function
- Modular exponentiation
- Discrete logarithm problem

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RSA Cryptosystem Overview



- Invented by Rivest, Shamir, Adleman
- Based on integer factorization problem
- Security relies on difficulty of factoring n=pq

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RSA Key Generation Example

- Choose p=61, q=53
- Compute n=pq=3233
- Compute phi(n)=3120
- Choose e=17
- Compute d=2753

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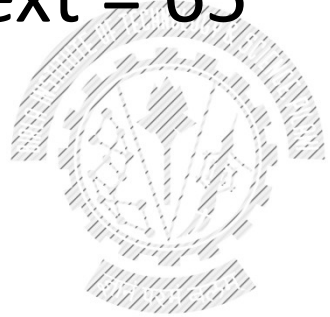
RSA Encryption Example

- Plaintext m=65
- Ciphertext c=m^e mod n
- c=2790

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RSA Decryption Example

- m=c^d mod n
- Recovered plaintext = 65



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ElGamal Cryptosystem Overview

- Based on discrete logarithm problem
- Encryption uses randomness
- Provides semantic security

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ElGamal Example

- p=23, g=5, x=6
- Public key y=8
- Encryption produces pair (c1,c2)
- Decryption recovers message

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Diffie-Hellman Key Exchange

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- Allows secure key agreement
- Uses modular exponentiation
- Shared secret computed independently

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Diffie-Hellman Example

- p=23, g=5
- Alice private key=6, public key=8
- Bob private key=15, public key=19
- Shared secret=2

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Security Assumptions

- RSA based on factoring hardness
- ElGamal based on discrete log hardness
- Diffie-Hellman based on discrete log hardness

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Applications

- Secure web communication (TLS)
- Digital signatures
- Secure email
- Key exchange protocols

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Summary

- Public key cryptography solves key distribution
- RSA, ElGamal, Diffie-Hellman are fundamental
- Used in modern secure systems

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